



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT407 **COURSE TITLE:** Internet of Things and Application

UNITS: 3.0

TIME ALLOWED: 2 Hours

Instruction: Read Carefully and Answer any Three (3) questions. Each question is 20 marks.

1. (a) List IoT characteristics and state the benefits of adopting IoT based technology to oil and gas.
1. (b) Explain the different levels of IoT with diagrams
1. (c) Develop a physical design for agricultural applications on an appropriate level. State the benefits of the chosen level. 1. 2.
2. (a) Distinguish between IoT and M2M.
2. (b) Discuss with a clear diagram IoT communication models.
2. (c) List IoT communication technologies and discuss any four
3. (a) Explain with a clear diagram SDN *SDN Defined Net*
3. (b) Compare SDN with traditional network
3. (c)(i) Explain cloud computing with different services
3. (c)(ii) What are the features of big data.
4. (a)(i) Compare OSI model to TCP/IP model
4. (a)(ii) Discuss with reference to the OSI model the following: Switch; Routers and Gateways
4. (b) Explain IoT architecture with a block diagram, include appropriate protocol for each layer
4. (c) Discuss the 7-layer IoT reference model
5. (a) What is a WSN and list its characteristics.
5. (b) Discuss IoT design methodology
5. (c) Draw the basic building block of an IoT. Hence develop an IoT system for a smart home indicate the level with reason(s)

*Digital Block
Comm. model.
Comm. API*